

Supporting Information

Occurrence and fate of perfluorinated compounds in soil following land application of biosolids

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Table S1 PFCs and surrogate standards used in this study

Analyte	Surrogate Standard
PFBA	[¹³ C ₄] PFBA
PFPeA	[¹³ C ₄] PFBA
PFHxA	[¹³ C ₂] PFHxA
PFHpA	[¹³ C ₂] PFHxA
PFOA	[¹³ C ₄] PFOA
PFNA	[¹³ C ₅] PFNA
PFDA	[¹³ C ₂] PFDA
PFUnA	[¹³ C ₂] PFUnA
PFDoA	[¹³ C ₂] PFDoA
MeFOSAA	[² D ₃] N-MeFOSAA
EtFOSAA	[² D ₅] N-EtFOSAA
PFBS	[¹⁸ O ₂] PFHxS
PFHxS	[¹⁸ O ₂] PFHxS
PFHpS	[¹⁸ O ₂] PFHxS
PFOS	[¹³ C ₄] PFOS
PFDS	[¹³ C ₄] PFOS

Desorption Experiments. For the equilibrium experiments, three sets of triplicate reactors were prepared for each soil type by transferring soil and synthetic fresh water [1] with 1 g/L of sodium azide to a 50 mL polypropylene tube to achieve a 1:10 soil-water ratio (r_{sw} , Soils W and K) or a 1:20 r_{sw} (Field 2). These r_{sw} values were selected for each soil to ensure aqueous PFC concentrations would remain within the standard linear calibration curve for direct injection aqueous analysis. Reactors were vortexed and placed on a shaker table for the duration of the experiment. For the equilibrium desorption study, one reactor from each triplicate set was sampled at each time point such that each reactor was sampled every fourth time step. Three triplicate sets of reactors were prepared for each soil to provide triplicates for each sampling time point. Prior to sampling, the vials were centrifuged at 2,700 rpm for 20 min, a 675 μ L sample of the aqueous phase removed, and a replacement aliquot of 675 μ L of clean synthetic freshwater was added to each reactor. Reactors for Soil W and Soil K were sampled at 1 hr, 2 hr, 6 hr, 12 hr, 24 hr, and then every subsequent 24 hrs for 14 days. Reactors for Field 40 were sampled at 15 min, 30 min, 45 min, 1 hr, 6 hr, 12 hr, 24 hr, and then every subsequent 48 hrs for 14 days. All reactors were sampled on day 14. At the completion of the 14-day desorption equilibrium experiments, the aqueous phase was removed and the solid phase from a subset of reactors was extracted to determine residual fraction of PFCs in the solid phase and to enable the completion of mass balance calculations. (Table S5)

Soil and Biosolids Extractions. Moisture content of each sample was separately measured by drying overnight in an oven at 105°C to enable reporting of the results on an oven-dry dry weight basis. Samples were weighed into 50 mL polypropylene conical tubes. An aliquot containing 2 ng of each surrogate standard was added to each tube immediately prior to extraction. A solvent mixture of 99:1 (v/v) methanol and ammonium hydroxide was employed for each extraction.

Seven mL of the solvent mixture was added to each tube prior to vortexing and placement of each tube in a heated (30°C) sonication bath (Fisher Scientific FS110H, Pittsburg, PA) for one hour. Each tube was subsequently removed and placed on a shaker table (VWR 5000 STD 120V, West Chester, PA) for two hours. The samples were then centrifuged (Eppendorf 5810, Hamburg, Germany) at 2,700 rpm for 20 minutes (1467 RCF), and the extract was decanted into a clean, glass 20-mL scintillation vial. This procedure was repeated twice for a total of three extraction cycles. The combined extract was then evaporated to dryness under nitrogen (Organomation Associates Inc. N-EVAP 112, Berlin, MA) and reconstituted with 700 µL of 99:1 (v/v) methanol and acetic acid. The reconstituted extract was transferred to a microcentrifuge tube containing 20-40 mg of ENVI-Carb (Supelco Inc., Bellefonte, PA), vortexed by hand for thirty seconds, and centrifuged (Fisher Scientific, accuSpin Micro 17) at 17,000 RCF for 30 minutes. An aliquot of each cleaned extract (105 µL) was transferred to an autosampler vial containing 1,350 µL of Milli-Q water (Millipore, Billerica, MA) and 45 µL of dilution water (0.01% ammonium hydroxide in Milli-Q water). Autosampler vials were then vortexed by hand for thirty seconds prior to analysis by liquid chromatography tandem mass spectrometry (LC-MS/MS).

Aqueous Sample Preparation. Aqueous phase samples from batch desorption experiments were prepared by transferring 675 µL of aqueous sample into a microcentrifuge tube with 675 µL of optima methanol and a 150 µL aliquot of 70/30 solution containing 0.3 ng of each surrogate standard. Microcentrifuge tubes were vortexed for thirty seconds and centrifuged at 17,000 RCF for 30 minutes. A volume of 1,300 µL of the prepared sample was then transferred to an autosampler vial for direct injection LC-MS/MS analysis.

LC-MS/MS Analysis. Chromatography was performed using an aqueous ammonium acetate (10 mM) and methanol (10mM) gradient delivered at a flow rate of 800 µL/min by an Agilent LC system (Agilent 1200) Binary pump controlled by an Agilent 1200 series Instant Pilot controller (Santa Clara, CA). Samples and standards were injected (1 mL) by a CTC Analytics LEAP Technologies autosampler onto a 50 mm x 4.6 mm Gemini C18 column (3-micron particle size, Phenomenex, Torrance, CA) equipped with a C18 Guard Column and Guard Cartridge (Higgins Analytical). For analysis of soil extracts, initial eluent conditions were 50% methanol and 50% water. The percent methanol was ramped to 95% over 4 min, held at 95% over 4 min, ramped down over 1.5 min, and held at 50% for 2.5 min. For analysis of aqueous samples, initial eluent conditions were 5% methanol and 95% water. The percent methanol was ramped to 60% over 0.75 min, ramped to 100% over 4 min, held at 100% over 3 min, ramped down to 5% over 0.5 min, and held at 5% for 1.75 min. A MDS Sciex Applied Biosystems 3200 Q trap mass spectrometer (MDS Sciex, Ontario) operating in negative electrospray ionization scheduled multiple reaction monitoring (MRM) mode was employed for sample analysis. Two MRM transitions were acquired for all analytes. The monitored transitions were analyte dependent and are the same or similar to those used previously [2]. For PFCAs, these transitions correspond to decarboxylation transitions, while the monitored PFAS transitions were the formation of the FSO_3^- ion ($m/z = 99$). Quantitation was performed using Analyst®, with all calibration curves having r^2 values greater than 0.99 and the accuracy of each calibration point within 30% of its expected value.

Optimal instrumental source parameters were determined and are as follows: ionspray voltage - 4,000 V, curtain gas flow 35 arbitrary units (au), nebulizer gas flow 60 au, turbo gas flow 50 au, medium collision gas flow, and source temperature 650 °C. Zero air provided by a Parker-Balston Source 5000 Zero Air Generator (Haverhill, MA) was used for the nebulizer and drier gas, and nitrogen was used as the curtain and collision gas

Quality Assurance/Quality Control Measures. To account for variations in the samples, all soils and biosolids were extracted in triplicate and all desorption reactors were prepared in sets of three. In all cases, analytical results are reported as the average of the three values. If one of the three values was below detection then the average value was reported as below detection.

To determine the effectiveness of the soil/biosolid extraction method, soils and biosolids were spiked with a known amount of PFC stock standard solution (Table S2). Similarly, spike recovery experiments were conducted with aqueous phase samples to determine any potential loss of PFCs during sample preparation. To further ensure that three extractions were sufficient to recover PFCs from soils and biosolids, triplicate samples from each matrix were subjected to four sequential extractions, with each extract analyzed separately. The amount of mass recovered in the fourth extraction was less than 5% in all cases except PFHxA (8%) in biosolids (Figure S1). To determine the effectiveness of the soil/biosolid extraction method, soils and biosolids were spiked with a known amount of PFC stock standard solution. The extraction method recovered 69% (PFNA) to 153% (PFDS) of PFCs (Table S1).

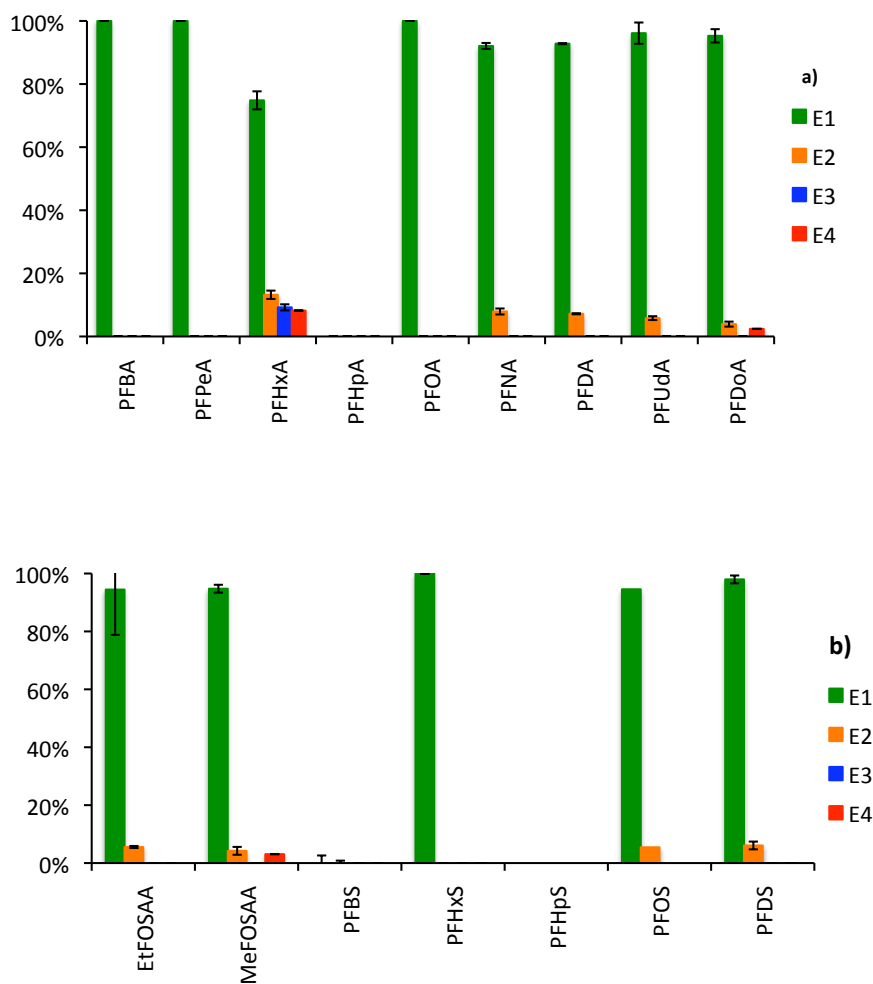
Table S2 Biosolids and biosolids amended soil spike recovery results

Analyte	% Recovery		% Recovery	
	from Biosolid	STDEV	from Soil	STDEV
PFBA	92%	3%	101%	6%
PFPeA	125%	4%	115%	11%
PFHxA	99%	2%	105%	7%
PFHpA	101%	3%	99%	10%
PFOA	101%	3%	114%	3%
PFNA	69%	1%	73%	3%
PFDA	84%	3%	89%	3%
PFUdA	100%	1%	101%	10%
PFDoA	95%	2%	101%	10%
EtFOSAA	101%	8%	98%	14%
MeFOSAA	108%	12%	118%	13%
PFBS	144%	3%	131%	8%
PFHxS	104%	1%	102%	8%
PFHpS	118%	3%	117%	8%
PFOS	95%	3%	85%	6%
PFDS	153%	35%	96%	4%

Similarly, spike recovery experiments were conducted with aqueous phase samples to determine any potential loss of PFCs during sample preparation. Synthetic fresh water spiked with a known amount of PFC stock standard solution and PFC internal standard solution was placed in either 50 mL polypropylene tubes or microcentrifuge tubes and subsequently transferred to autosampler vials to test for losses to each type of container. Recoveries from 50 mL tubes were 77-95%, with the exceptions of the longer chain length PFCAs (PFUdA and PFDoA) and the PFOS precursors (MeFOSAA and EtFOSAA), for which the recoveries were 32-58%. Additionally, over-recovery of

PFBA (174%) and PFPeA (271%) was evident, reflecting an unsatisfactory analytical bias for these PFCs. As a result, aqueous-phase data for PFBA, PFPeA, PFUdA, PFDoA, MeFOSAA, and EtFOSAA were removed from additional analysis. Recoveries of all other analytes from spiked microcentrifuge tubes were 85-103%.

Finally, control reactors were prepared for desorption experiments by preparing synthetic freshwater with a known amount of the PFC stock standard solution and PFC surrogate standard solution in a 50 mL polypropylene tube with no solid phase. These reactors were placed on the shaker table sampled on the final day. If losses for an analyte were greater than 10%, then these results were used to calculate mv for the analyte (see Equation 1) to account for losses to the tube when determining K_{des} values.



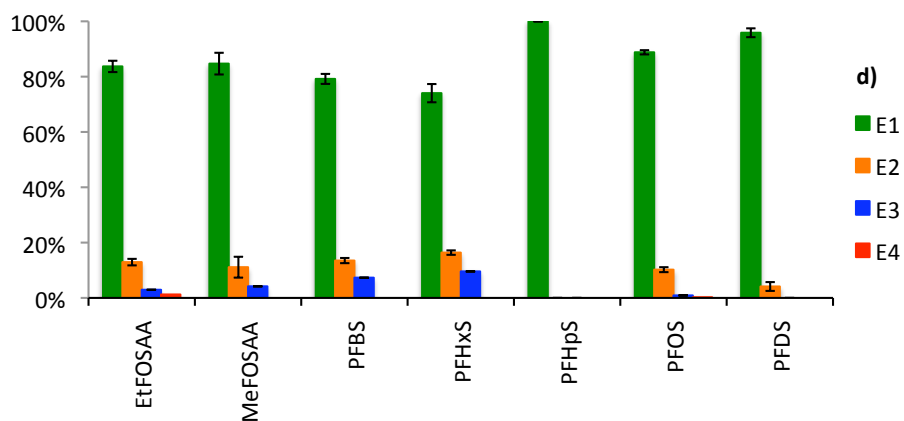
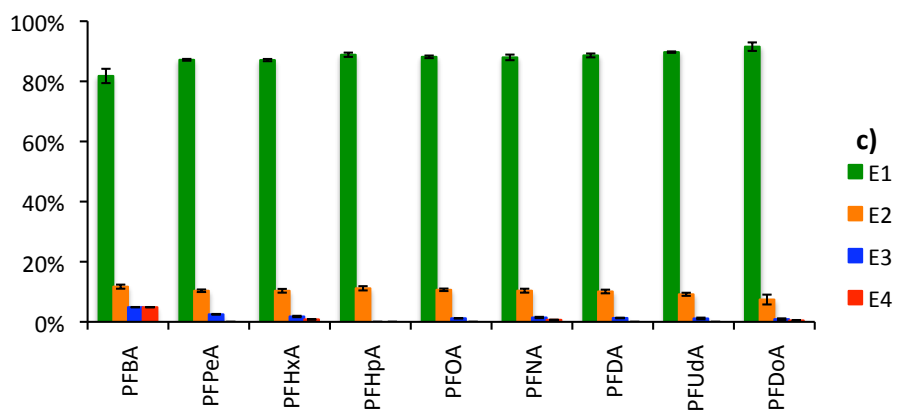
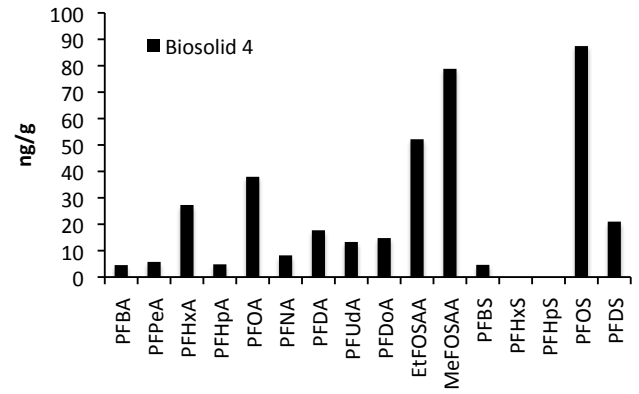
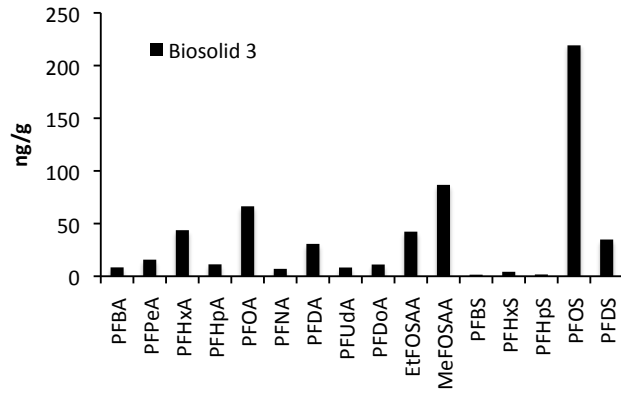
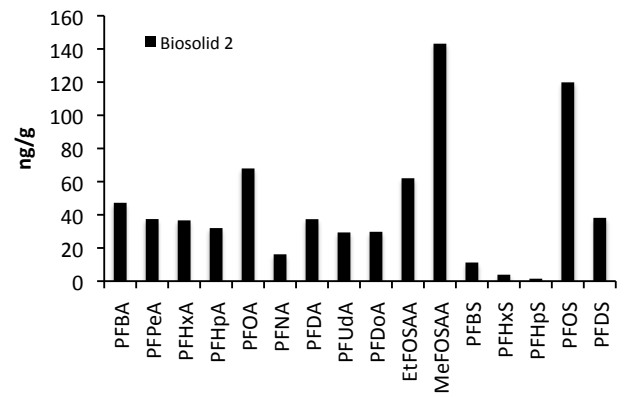
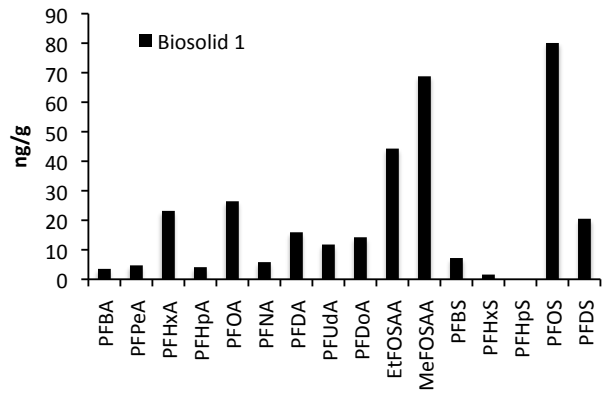


Figure S1. Results of extraction efficiency experiments in biosolids (a-b) and soils (c-d). Reflects the percentage of each PFC recovered in each extraction in a sequence of four extractions. E1=extraction 1, E2=extraction 2, E3=extraction 3, and E4=extraction 4.



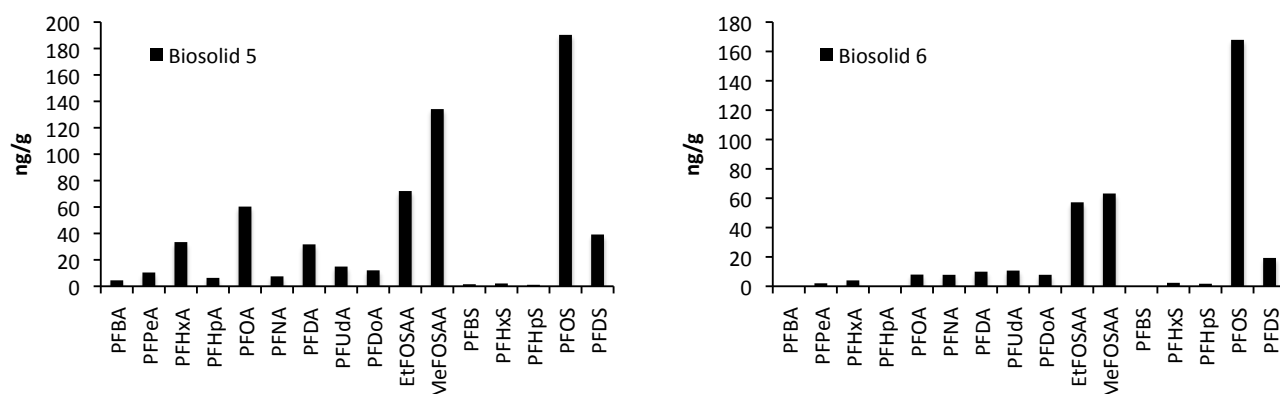


Figure S2. Occurrence of PFCs in six source Biosolids for short-term W and K Soils.

Table S3 Results of linear regressions of soil concentrations with loading rate.

Analyte	Slope	±	Y-intercept ^a	r ²
PFBA	0.001	0.0001	0	0.65*
PFPeA	0.003	0.0003	0	0.86*
PFHxA	0.003	0.0003	0	0.84*
PFHpA	0.003	0.0002	0	0.92*
PFOA	0.015	0.0009	0	0.94*
PFNA	0.003	0.0002	0	0.94*
PFDA	0.009	0.0004	0	0.97*
PFUdA	0.003	0.0002	0	0.95*
PFDoA	0.006	0.0003	0	0.96*
EtFOSAA	0.018	0.0013	0	0.92*
MeFOSAA	0.018	0.0013	0	0.86*
PFBS	0.000	0.0003	0	0.26
PFHxS	0.004	0.0003	0	0.94*
PFHpS	0.002	0.0001	0	0.96*
PFOS	0.198	0.0153	0	0.90*
PFDS	0.044	0.0020	0	0.96*

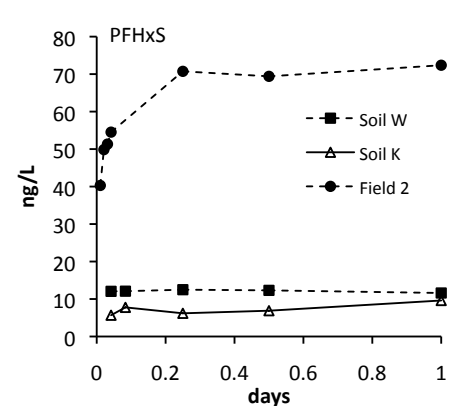
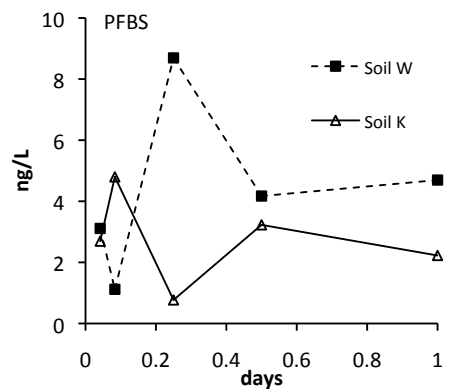
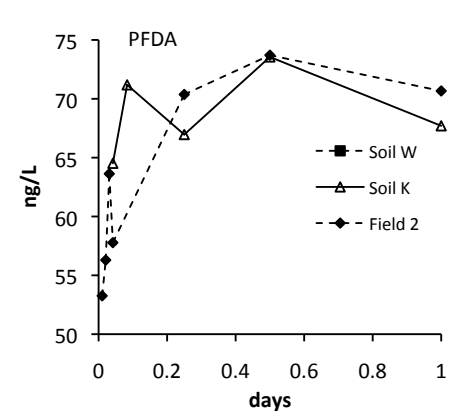
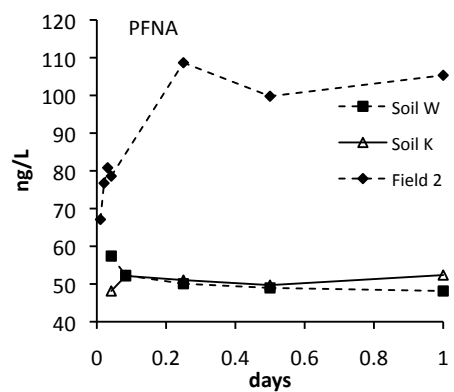
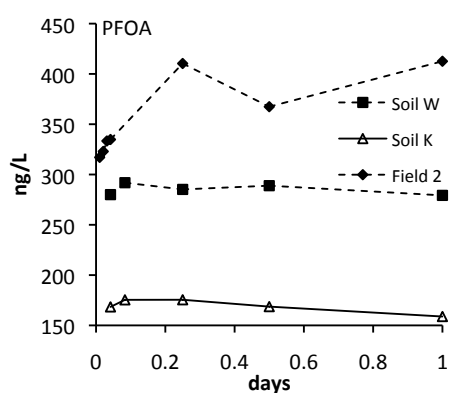
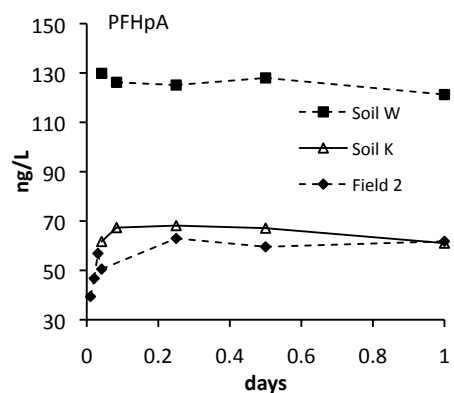
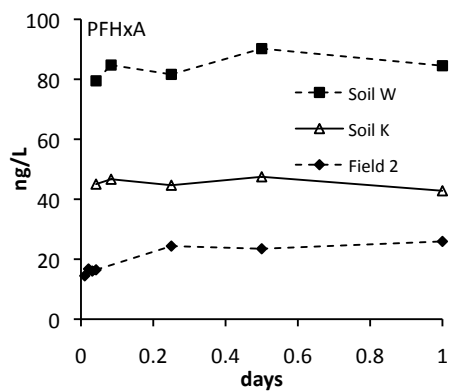
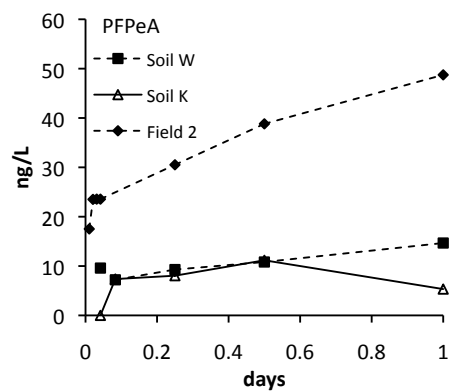
^aY-intercepts for all cumulative regressions were forced to equal zero because PFC concentrations in plots receiving no biosolids are assumed to be zero. *Indicates a significant (p<0.05) relationship between loading rate and PFC concentration.

Standard Additions. This study employed the PFHxS stable isotope as a surrogate standard for PFBS (Table S1). The use of PFHxS as a surrogate standard for PFBS may contribute to potential enhanced recovery of PFBS in biosolids amended soils due to differential matrix effects in the soils and biosolids. To evaluate potential matrix effects, standard additions for PFBS were performed in Soils W and K at the highest loading rates and in the source biosolid for each soil that had the lowest measured PFBS concentration.

Standard additions showed no difference in instrument response for PFBS calibrations performed in solvent, soil extracts, or biosolids extracts. However, instrument response for PFHxS differed between the matrices, with additional signal suppression of PFHxS in soil extracts when compared to biosolids extracts. This indicates that matrix effects were different between PFBS and PFHxS and also between soils and biosolids, potentially leading to analytical bias. Because of the use of PFHxS as a surrogate standard for PFBS, the suppressed recovery of PFHxS in soil indicates that measured PFBS soil concentrations are slightly enhanced. Results confirm this, as the concentrations of PFBS in the subset of soil samples (determined directly using PFHxS as the surrogate standard) were approximately 1.5-2.5 times higher than those obtained by standard addition. Using the PFBS soil concentrations determined by standard addition, mass balance recoveries of PFBS in Soils W and K were recalculated for the two highest loading rates. In Soil W, recovery of PFBS was adjusted from 450% to 185% with use of standard addition PFBS concentrations, and PFBS recovery in Soil K was adjusted from 2500% to 1600%. These data suggest that while matrix effects are partially responsible for the higher PFBS recovery observed in the soils, there were still significant increases in PFBS in the soils when compared to what was applied via the biosolids.

Table S4 Mass balance results of actual vs. calculated soil PFC concentrations

Analyte	Soil W		Soil K	
	% Recovery	±	% Recovery	±
PFBA	51%	11%	151%	14%
PFPeA	152%	43%	282%	38%
PFHxA	69%	18%	48%	18%
PFHpA	139%	39%	282%	39%
PFOA	110%	10%	125%	17%
PFNA	118%	29%	231%	93%
PFDA	125%	38%	301%	39%
PFUdA	78%	32%	169%	20%
PFDoA	115%	52%	316%	28%
PFTTrDA	<LOQ	<LOQ	128%	22%
EtFOSAA	64%	5%	69%	12%
MeFOSAA	23%	4%	69%	15%
PFBS	322%	104%	2228%	365%
PFHxS	99%	13%	97%	70%
PFHpS	103%	67%	96%	N/A
PFOS	97%	39%	86%	26%
PFDS	97%	44%	44%	25%



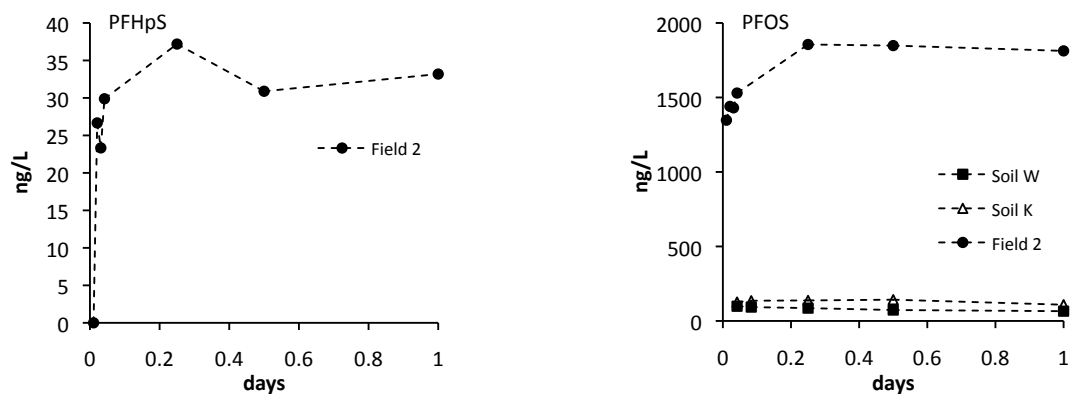


Figure S3. Desorption equilibrium of PFCs from Soil W, Soil K, and Field 2. If a soil is not included in a plot, then this indicates that aqueous concentrations for that particular PFC were below detection in equilibrium experiments with the excluded soil. If a PFC is not included, then it was below detection for equilibrium experiments with all soils. Data beyond 24 hrs are not shown as the slope of the measured aqueous concentrations vs. time was not statistically different ($p < 0.05$) from zero after the initial 24 hrs.

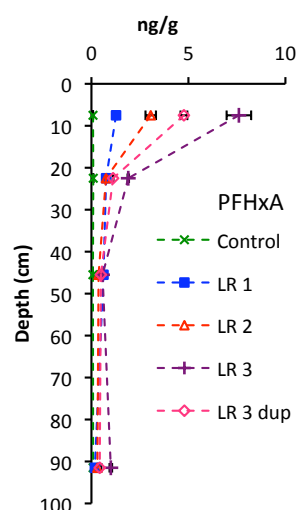
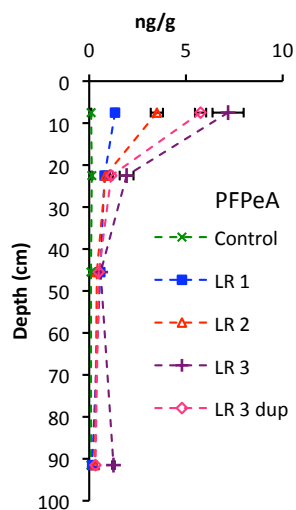
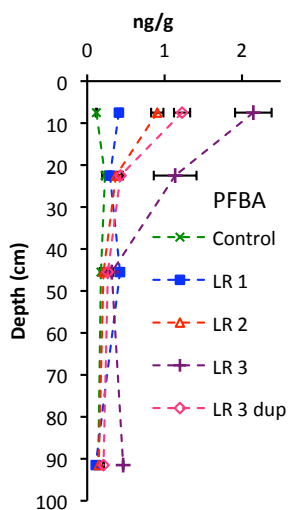
Table S5 Total aqueous plus total solid phase recovery of PFCs in equilibrium desorption reactors.

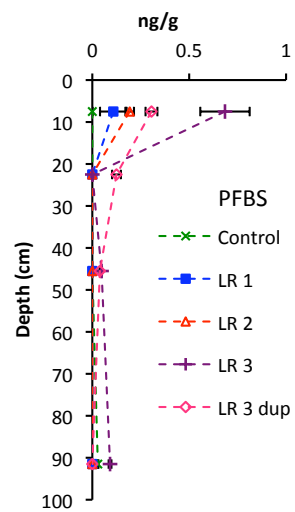
Analyte	Recovery		Recovery		Recovery	
	Soil W	STDEV	Soil K	STDEV	Field 2	STDEV
PFHxA	100%	5%	106%	11%	123%	25%
PFHpA	82%	4%	85%	11%	128%	19%
PFOA	103%	9%	100%	5%	86%	9%
PFNA	100%	8%	105%	9%	105%	3%
PFDA	93%	6%	85%	1%	100%	8%
PFBS	110%	21%	97%	30%	N/A	N/A
PFHxS	65%	6%	N/A	N/A	84%	8%
PFHpS	N/A	N/A	N/A	N/A	102%	6%
PFOS	113%	8%	102%	7%	91%	5%
PFDS	N/A	N/A	N/A	N/A	N/A	N/A

A value of N/A indicates that concentrations in the aqueous and/or solid phases were below detection and therefore recovery could not be calculated.

Table S6 Individual and average K_{oc} results for Soils W, K, and General Fields

Analyte	Soil Type	Log K_{oc}	STDEV
PFHxA	Soil W	1.63	0.52
	Soil K	2.35	0.55
	Gen Fields	1.74	0.37
	Average	1.91	0.39
PFHpA	Soil W	2.22	0.12
	Soil K	2.82	0.11
	Gen Fields	1.52	0.73
	Average	2.19	0.65
PFOA	Soil W	2.42	0.24
	Soil K	2.59	0.21
	Gen Fields	1.92	0.31
	Average	2.31	0.35
PFNA	Soil W	2.42	0.05
	Soil K	2.59	0.12
	Gen Fields	1.99	0.24
	Average	2.33	0.31
PFDA	Soil W	3.14	0.07
	Soil K	3.32	0.05
	Gen Fields	3.05	0.07
	Average	3.17	0.14
PFHxS	Soil W	2.81	0.10
	Soil K	3.02	0.20
	Gen Fields	2.28	0.30
	Average	2.70	0.38
PFOS	Soil W	3.31	0.06
	Soil K	3.62	0.03
	Gen Fields	3.08	0.05
	Average	3.34	0.27





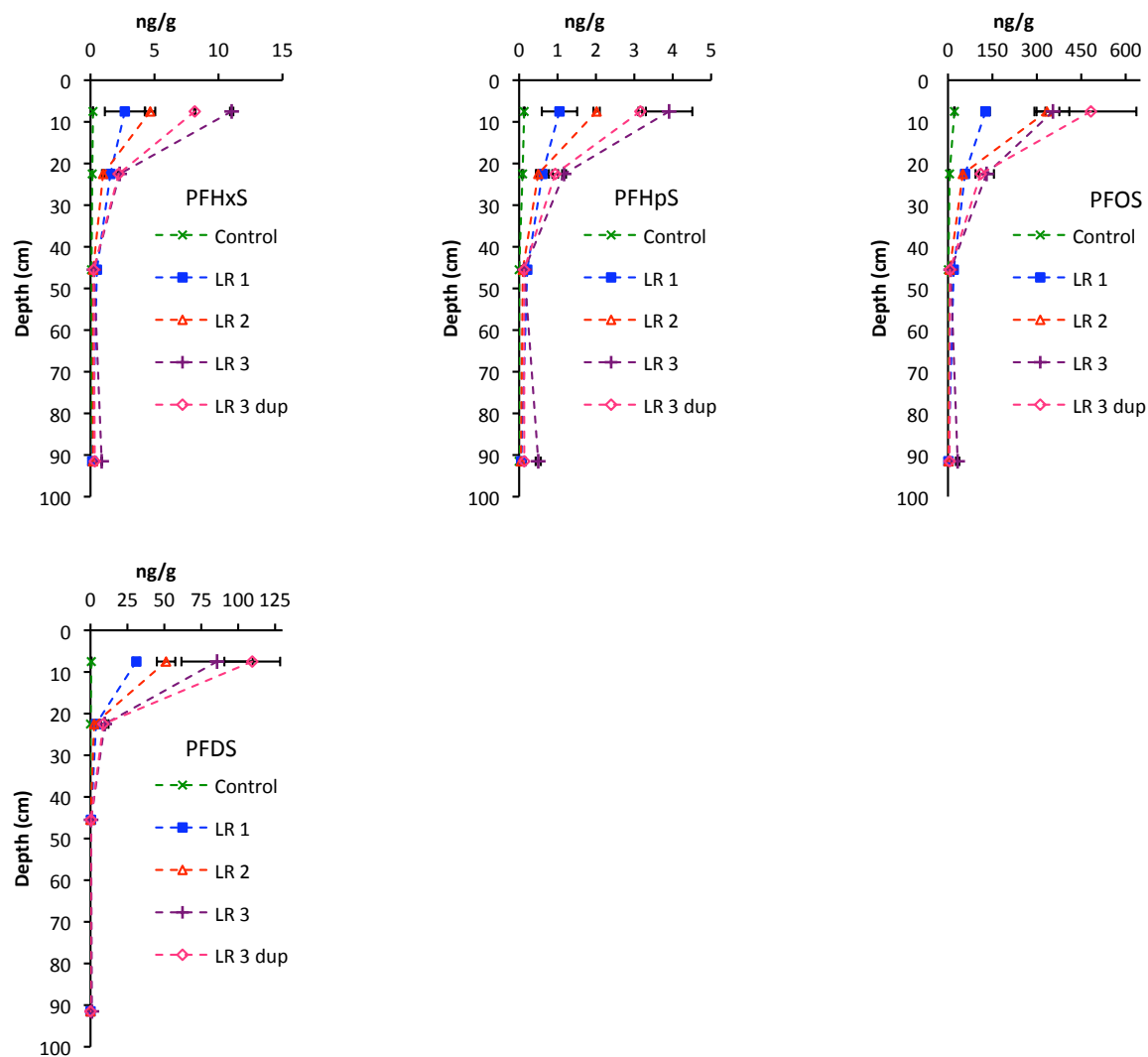


Figure S4. Concentrations of PFCs with depth in the long-term plots at various cumulative loading rates. Control = 0 Mg/ha, LR 1 = 553 Mg/ha, LR 2 = 1109 Mg/ha, and LR 3 and LR 3 dup = 2218 Mg/ha (on dry weight basis). In some instances, small increases in PFCs were measured between the 30-61 cm and 61-122 cm depth intervals, which could be simply due to preferential flow paths created as an artifact of excessive soil sampling because these plots have been sampled annually since 1973.

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- (1) *Methods for Measuring the Toxicity and Bioaccumulation of Sediment-associated Contaminants with Freshwater Invertebrates*; EPA 600/R-99/064; USEPA Office of Research and Development: Duluth, MN, 2000.
- (2) Higgins, C. P.; Field, J. A.; Criddle, C. S.; Luthy, R. G. Quantitative determination of perfluorochemicals in sediments and domestic sludge. *Environ. Sci. Technol.* **2005**, *39*, 3946-3956.